UNIVERSITY OF LIVERPOOL

SCHOOL OF ENGINEERING

MNFG604 Assessment 2

A Reverse Engineering Project

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Summary

The purpose of this project is to demonstrate skills acquired in technical modelling through the careful selection of an existing product with appropriate complexity and re-creating it as a PTC Creo Parametric assembly. The product selected for this project was an air freshener developed by S.C Johnson & son under the brand name Glade Sense and Spray Automatic Spray. This product fully fulfils a high level of complexity and contains the desired internal components such as an electric circuit, moving internal parts and batteries.

Table of contents

1. Introduction

The goal of this project is to reverse engineer a glade sense and spray automatic air freshener. It is a battery powered device equipped with a motion sensor that detects movement and releases a fragrance within the room. When motion is detected, and a shadow is cast on the sensor, this completes the circuit causing a motor within the device to activate thus turning gears to press down a plastic component within onto the fragrance capsule to release the spray.

The objective of this project is to deconstruct the existing device and accurately measure all critical dimensions and analyse its structural, mechanical and electrical elements, to reconstruct it and develop a detailed assembly in PTC Creo parametric software. The final model will illustrate both the static features and moving parts within the device, emphasising realistic textures to replicate the design features on the original product.

This report provides a systematic review of the reverse engineered spray highlighting dismantling and measuring of structural elements, 3D modelling of components of the device, assembly integration while maintaining functional relationship between designed components, and documentation in the form of 2D technical drawings and a bill of materials for manufacturing insights.

Figure 1. ***glade air freshener***

2. Reverse engineering

2.1. Initial assessment

The air freshener is fitted with two AA batteries and a fragrance capsule upon first inspection. The sensor operates using a sensor located on the front surface of the air fresher that detects motion when a shadow is cast on it. This triggers the circuit within the air fresher causing the motor to activate rhus driving the reduction gears in a sequence which lead to rotation of a sector shaped gear that converts rotational motion into linear motion of a plastic nozzle piece that presses down on the fragrance capsule leading to it releasing the gas trapped within before returning to its original position.

Additionally, the air freshener operates with a timing sequence that releases the fragrance after a given period. This is to ensure periodic sprays regardless of detecting motion, it will release said fragrance withing the room. This mechanism is integrated within the circuit to maintain a consistent room fragrance.

The main components of the air freshener are:

- Fragrance capsule
- Batteries
- Reduction gear mechanism
- Motor
- Electric circuit
- Frame made of ABS plastic.

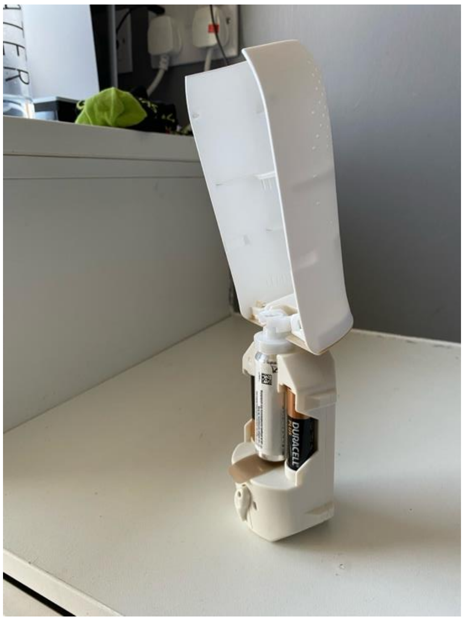

Figure 2: ***opened air freshener showing fragrance capsule and batteries.***

2.2. Dismantling

The top of the air freshener rotates along an axis by 180 degrees to show the fragrance capsule and batteries for easy replacement when empty. These are coincident to cylindrical shaped hollow sections in the top frame where they slot in with the plastic nozzle piece pressing down on the fragrance capsule. The overall all frame is made of ABS plastic with the top and bottom frame connected using four M3 x 5 mm screws.

Figure 3: image showing back of air freshener

Once removed, this exposes the reduction gear mechanism, sector shaped gear and plastic nozzle piece that presses on the fragrance capsule shown in figure 4 below. The gears are covered in grease to reduce friction resistance during period operation of the air freshener. When activated, the motor rotates the motor gear labelled, motor gear, which transfers motion to the reduction gear mechanisms composed of a 64 teeth gear and 56 teeth gear that transfers motion to the sector shaped gear with 17 teeth that converts rotational motion to linear motion of the plastic nozzle piece that applies pressure as it moves vertically on the fragrance capsule that releases the spray. The gears are attached to the internal frame of the air freshener with the holes of the gears coincident to protrusions on the frame.

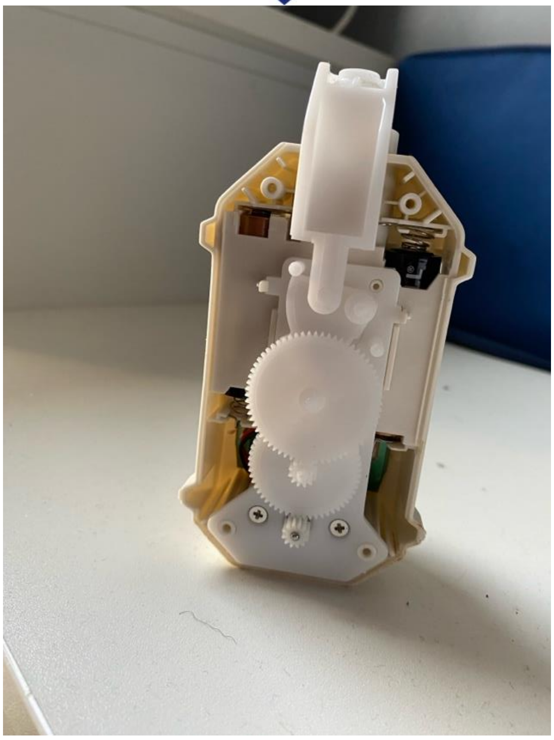

Figure 4: ***image showing reduction gear mechanism and internal components.***

The internal frame is attached to latches on the bottom frame whereby beneath it lies the motor and the electric circuit within the hollow space shown in figure 5 below. The motor is screwed to the internal frame by two M2.5 x 0.45 mm screws with the motor coincident to the frame with the electric circuit beneath it.

Figure 5: ***image showing back face with internal frame and motor removed.***

2.3. Modelling

The following functions in PTC Creo parametric were implemented to reverse engineer the components.

- Chamfer
- Extrude
- Revolve
- Surface revolve
- Chamfer
- Mirror part
- Round
- Sketch
- Datum planes
- Standard hole for screws
- Fillet

All modelled components were displayed using shading with edges under the graphics toolbar with different appearances set depending on the manufacturing material and original colour of the parts. Upon initial measurement, a great deal of care was taken to ensure that components fit together as intended therefore a hand drawn sketch was made and each components dimensions were compared to those that they were mated to before initial designs were made. Estimates on some dimensions were taken due to difficulty in accurately measuring different sections of the part. For more complex parts, sketches were made on one component and saved as a templet to import into a sketch for another part. This was done to ensure that parts align properly. Regarding more uniquely shaped parts such as the top cover, it was difficult to measure dimensions of the curved surfaces therefore rough estimates were taken and a simpler design was taken that deviates from the original component.

2.4. Parts and assemblies.

This section shows photographs of the individual components alongside the models recreated in PTC Creo parametric. Key measurements are presented to provide an overall idea of the size of each component.

2.4.1. Screws

There are two main types of screws within the air freshener with the four M3 x 5mm screws with overall length of 8 mm shown in figure 6a linking the bottom frame to the top frame through hollowed screw holes on the parts. The bottom frame has protrusions where the screws slot in.

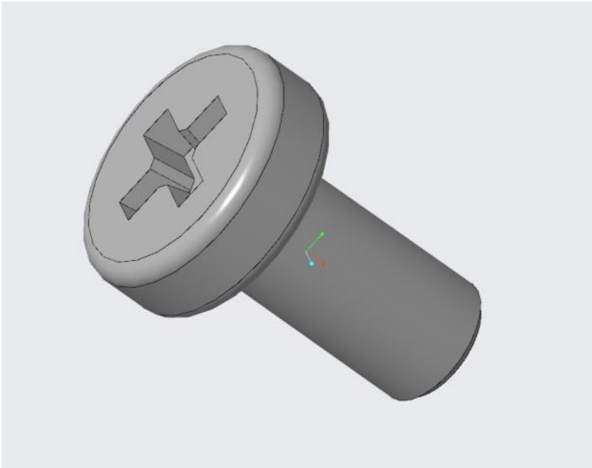

Figure 6a: ***images showing photograph of M3 screws to the left and model design to the right.***

The two M2.5 x 0.45mm screws with overall length 7 mm shown in figure 6b linking the motor to the internal frame within the air freshener through holes on the frame. Both screws are made of stainless steel with the threads along its surface represented on the model using cosmetic threads.

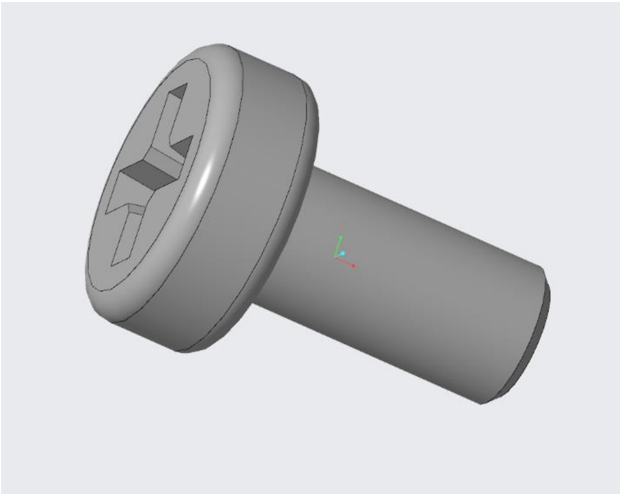

Figure 6b: ***images showing photograph of M2.5 screws to the left and model design to the right.***

2.4.2. Motor

The motor is made of aluminium with a radius of 15.8 mm and height of 24.6 mm including small protrusions at 2 mm on the bottom face where wires link it to the circuit board. The motor has 6 holes in a circular pattern on its top surface where the M2.5 mm screws slot in to link it to the internal frame and a hole where the pin sides in shown in figure 7 below.

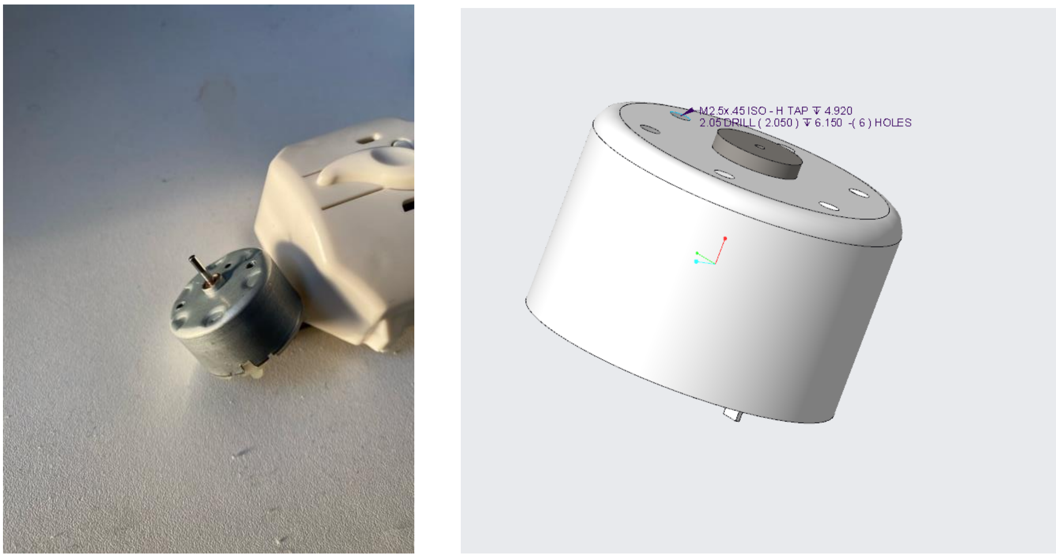

Figure 7: ***images showing photograph of motor to the left and model design to the right.***

2.4.3. Bottom frame

The bottom frame is made of ABS plastic with an overall length of 133.9 mm and width of 65.8 mm shown in figure 8 below. The frame is 1.9 mm thick with a height of 14.5 mm It has a top neck where the plastic nozzle piece lies with a height of 14.5 mm and 8.57 mm with two parallel plates 19.4 mm apart to ensure motion along a vertical axis. On the back face is a latch used to hang the air freshener in the room onto a hook attached to a surface and holes where the screws are connected through.

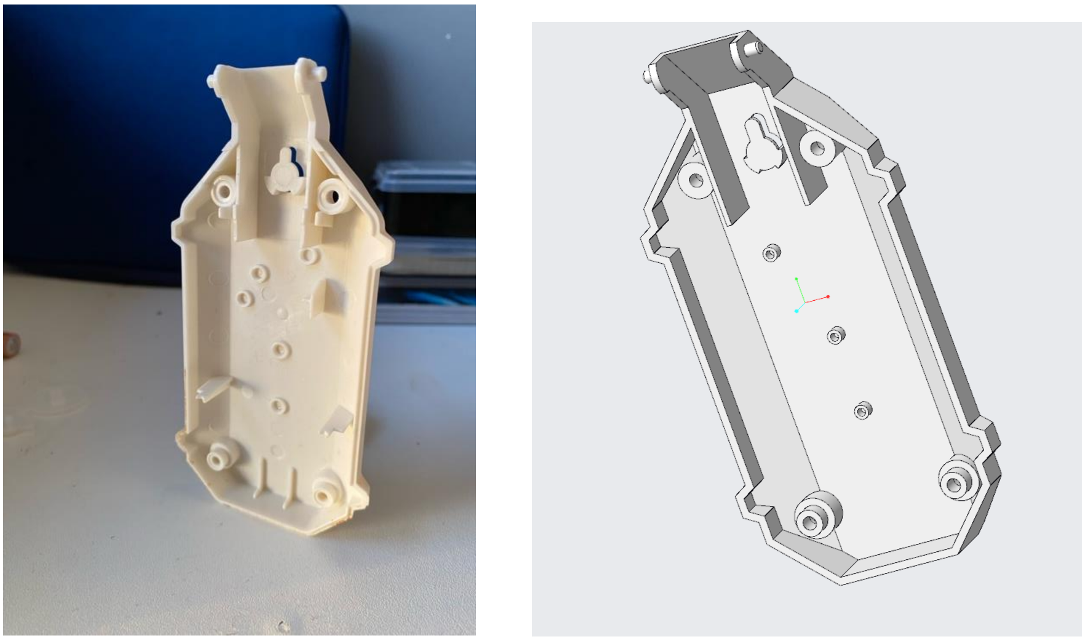

Figure 8: ***images showing photograph of back frame to the left and model design to the right.***

2.4.4. Top frame

The top frame is made of ABS plastic with an overall length of 113 mm and width of 65.8 mm. The frame is 1.9 mm thick with overall height of 44.24 mm. On its front surface are three cylindrical hollowed out regions where the fragrance capsule and two AA batteries lie parallel to each other. The height from base to the surface where the components lie is 19.3 mm whereby the other section is hollowed out for the motor and electric circuit board within the top frame. It has two parallel protrusions 10 mm apart and 1.5 mm thick that hold the internal frame in position. Additionally, there are cylindrical protrusions that hold the internal frame in place at radius 1.6 mm with two other cylindrical protrusions in the hollowed section where the M3 screws hold the top and bottom frames together. A button is designed on the from side of the hollow section that activates the circuit when required as well as holding the motion sensors.

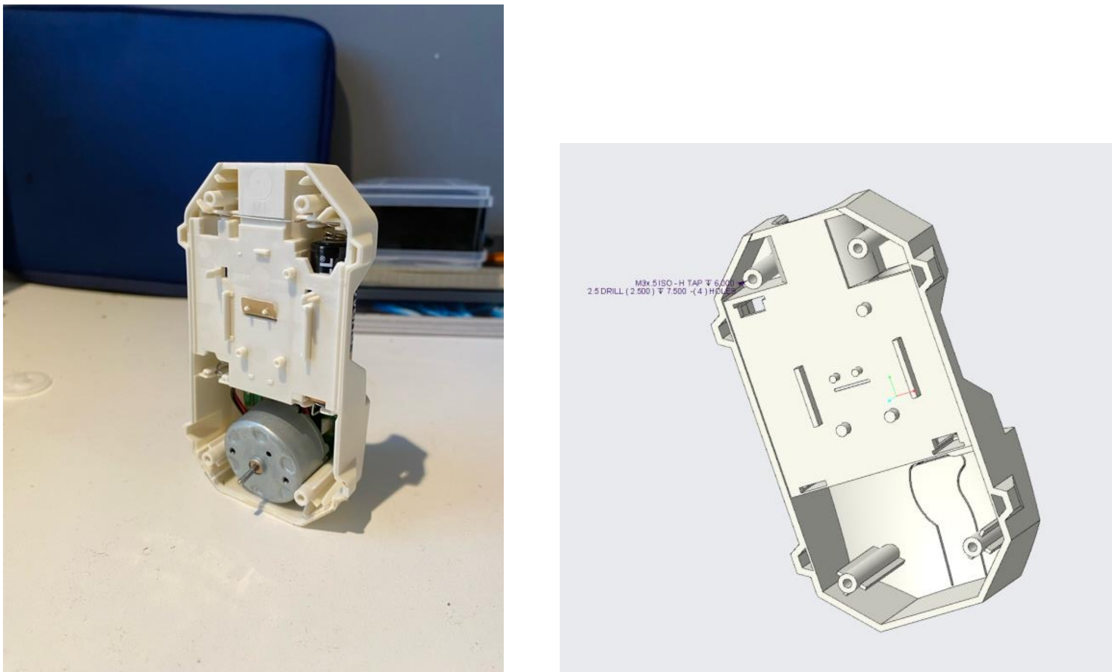

Figure 9: ***image showing photograph of back surface of top frame to the left and model design to the right***

2.4.5. Internal frame

The internal frame is made of ABS plastic with overall length of 84.5 mm and width of 49.96 mm shown in figure 10 below. the frame is 3.2 mm thick with a neck 60 mm in length and 28.41 mm in width. it has a hole in the lower section 4.75 mm in radius where the motor is positioned with protrusions on its front surface at height

8.6 mm and radius 1 mm that hold the gears in position on the frame as well as limit the rotation of the sector shaped gear to ensure vertical movement of the plastic nozzle piece. Additionally, it has two parallel holes with radius 2.7 mm where the M2.5 screws pass to link the motor to the internal frame.

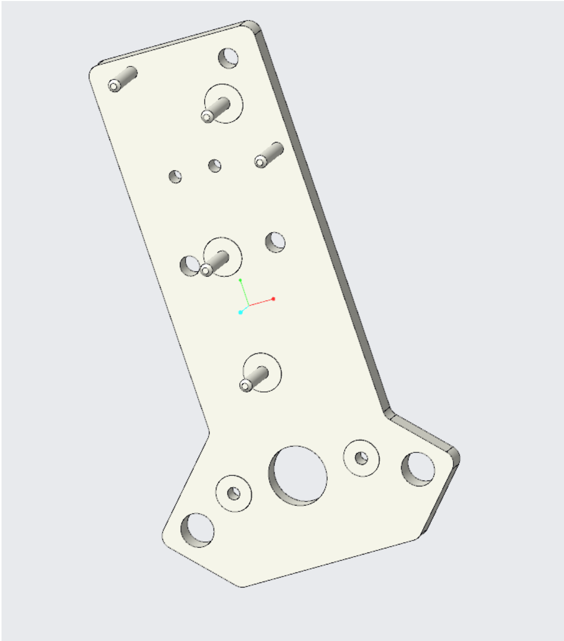

Figure 10: ***image showing photograph of internal frame to the right and model design to the left.***

2.4.6. Gears

The four gears are made of ABS plastic consisting of a 64 teeth gear, 56 teeth gear, sector shaped gear, and a 12 teeth motor gear. The 64 teeth gear has radius of 16.5 mm with overall height 6.5 mm shown in figure 11a. it has a lower section gear of height 3.8 mm and radius 3.5 mm with a 1 mm radius hole in the centre. The 56 teeth gear has radius of 14.5 mm with overall height of 6.5 mm shown in figure 11b. it has a lower section gear of height 3.8 mm and radius 3.5 mm with a 1 mm radius hole in the centre that holds it to the internal frame. The sector shaped gear has radius 22 mm with overall height 6.15 mm shown in figure 11c with angle between both surface at 75 degrees. It has indentations on both sides which limit its rotation through protrusions on the internal frame. The 12 teeth motor gear has overall radius 3.5 mm with overall height 6.5 mm shown in figure 11d. it has a 1.1 mm hole in the centre where motor pin links it to the motor.

Figure 11a: ***photograph and model design of 64 teeth gear***

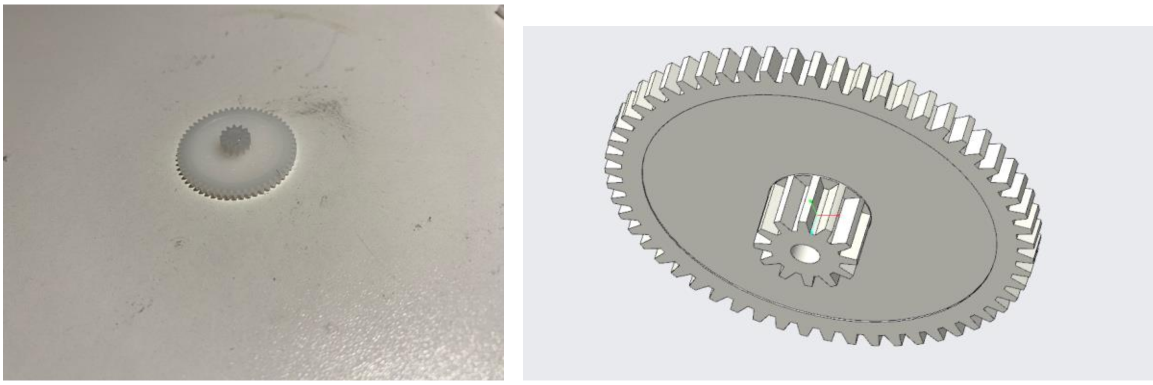

Figure 11b: ***photograph and model design of 56 teeth gear***

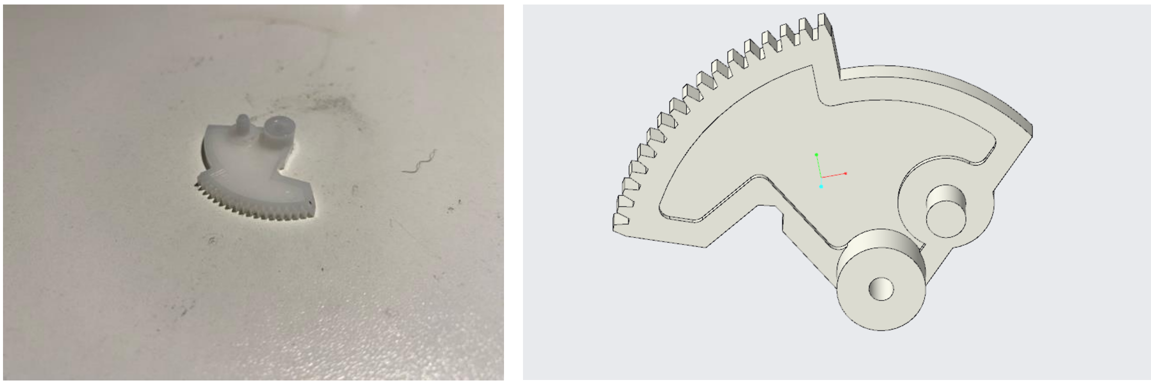

Figure 11c: ***photograph and model design of sector shaped gear.***

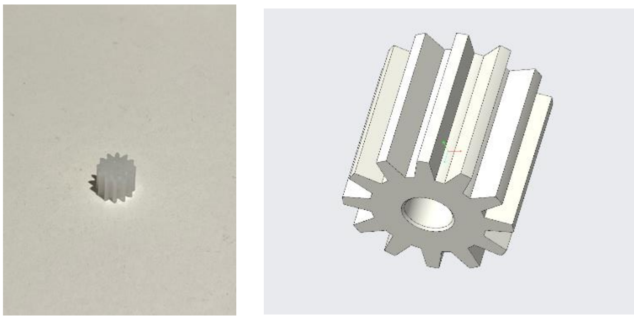

Figure 11d: ***photograph and model design of 12 teeth motor gear***

2.4.7. Plastic nozzle piece

The plastic nozzle piece is made of ABS plastic with overall length 35mm, height 54.75 mm and overall width 18.4 mm shown in figure 12. The piece is 2.8 mm thick and L-shaped with the top section 7.7 mm high and lower section 12.3 mm long. Its top section is angled at 45 degrees curved edges at a 3 mm radius. The top of the fragrance capsule is within the top section with a sphere at radius 1.8 mm. it has a 4 mm radius hole on the bottom section that attaches it to the top of the sector shaped gear.

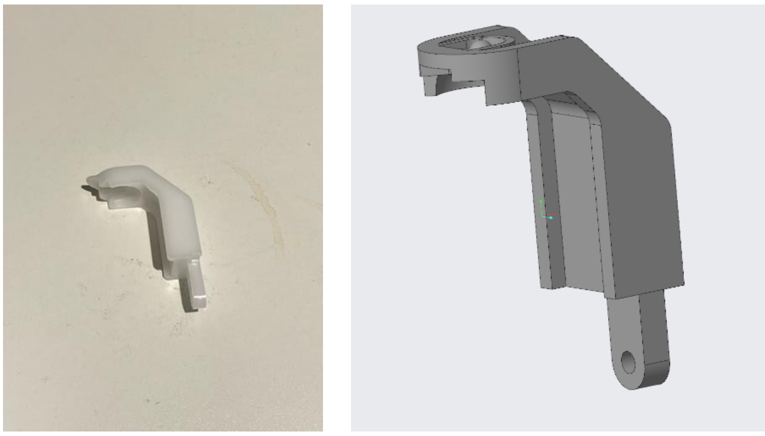

Figure 12: ***images showing photograph of plastic nozzle piece to the left and model design to the right.***

2.4.8. Fragrance capsule and capsule top

The fragrance capsule is made of polypropylene and has overall height of 76 mm and overall radius of 11 mm shown in figure 13. The top of the capsule has radius 5.26 mm and height 4.8 mm. It has a hole where the fragrance is accessed on the top section with radius 1.43 mm. The fragrance top is made of ABS plastic and has overall radius of 13.5 mm and height 8.8 mm with a thickness of 1 mm. It was difficult to separate the capsule top from the capsule hence they are presented together in the photograph below.

Figure 13: ***photograph of capsule with capsule top.***

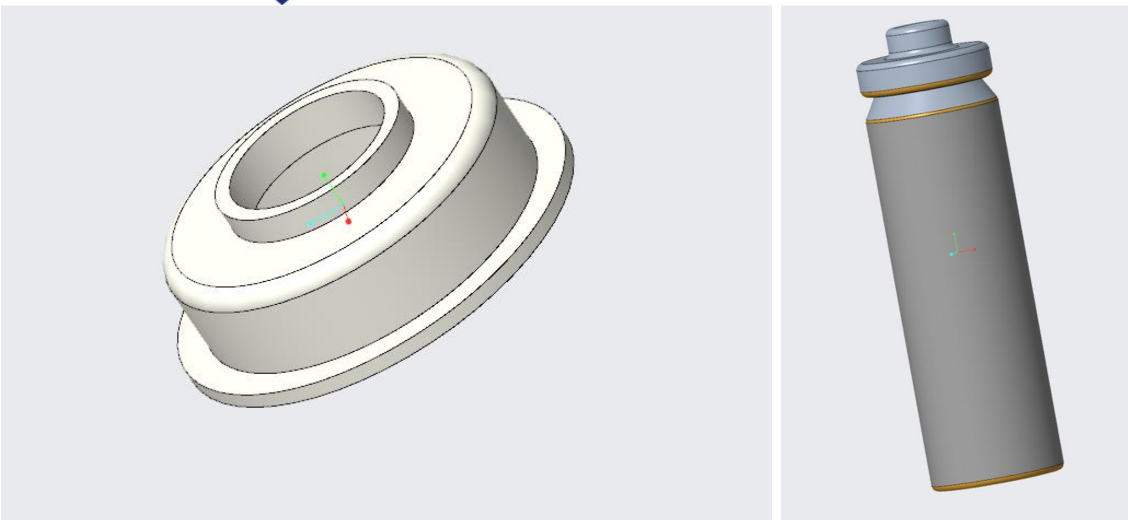

Figure 14: ***model design of capsule top to the left and capsule to the left.***

2.4.9. Batteries

The air freshener consists of 2 AA batteries made of polypropylene with overall height 49.7 mm and overall radius 6.8 mm shown in figure 15. The positive side of the battery has diameter 8.34 mm with the negative side 4.58 mm in diameter. The batteries lie on the top surface of the top frame parallel to each other with the fragrance capsule between them. There are holes on the top frame where the connection plates linking the batteries to the electric circuit board lie.

Figure 15: photograph of AA battery on the left and model design on the left.

2.4.10. Electric circuit

The electric circuit board is made from pre-peg material with overall length of 52.96 mm and overall height of 37.5 mm with thickness of 1 mm. The circuit board is shaped such that it fits within the hollowed section of the top frame with a 1 mm clearance between surfaces parallel to it shown in figure 16. The appearance of the circuit board design is sourced from images on the internet to replicate the layout of components on its surface.

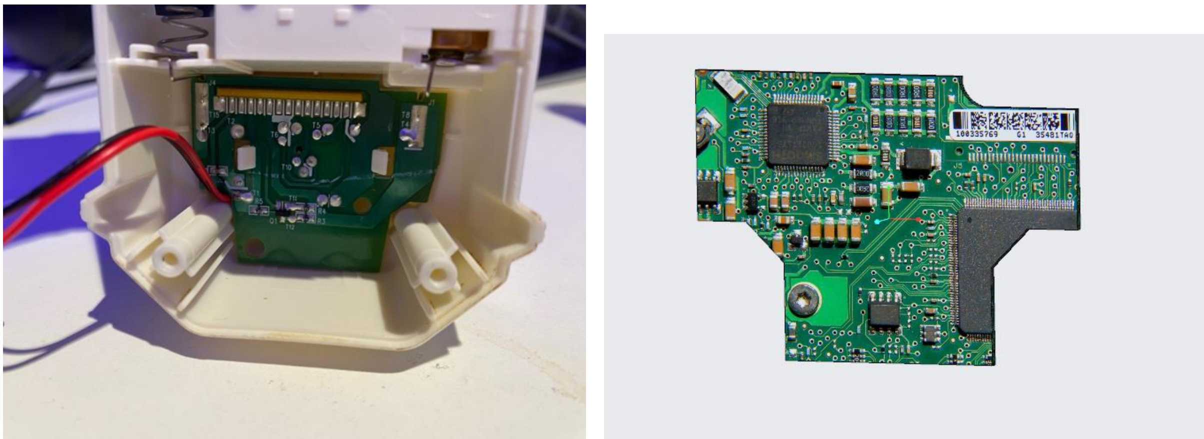

Figure 16: ***images of photograph of circuit board within top frame on the left and model design to the right.***

2.4.11. Top cover

The top cover is made of abs plastic with thickness of 1.5 mm and overall height of 143.3 mm. it has centred parallel plates 24 mm apart that attach it to the bottom frame and allow for rotation of the top cover to access the internal components of the air freshener shown in figure 22 below.

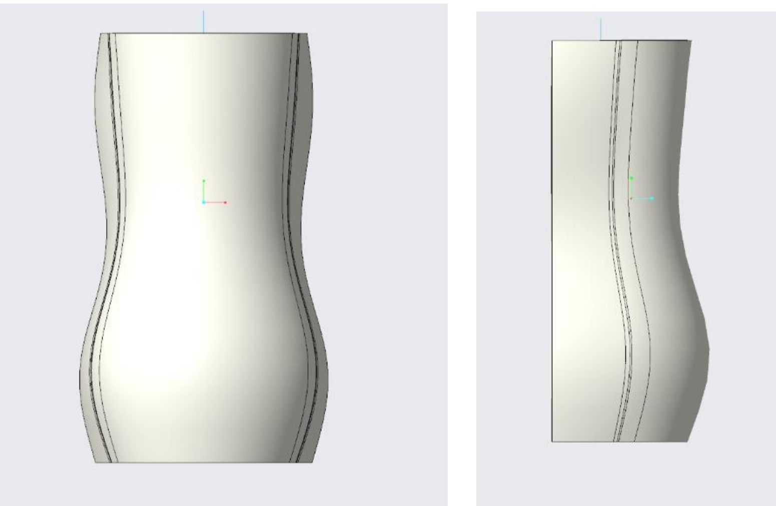

Figure 22: ***image showing model design of top cover from front and side face.***

2.5. Sub-assemblies

This section is dedicated to subassemblies of features in the final model to make mating each component to each other easier. It consists of the pin assembly, mechanism assembly with gears and the motor, the top assembly with the fragrance capsule and batteries, and the semi-complete assembly excluding the top cover.

2.5.1. Pin assembly

This assembly consists of the pin and gear pin that is mated to the motor to drive the gear reduction mechanism of the air freshener. The gear pin is set as the default connection type with the gear pin connection type set as pin to allow rotation along the pin axis with the curved surfaces of the pin and internal hole of the pin gear as coincident with a translation of 2.2 mm between the bottom surface of the pin and bottom surface of the pin gear.

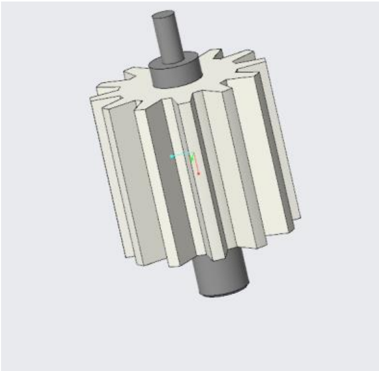

Figure 17: ***image showing assembly design of pin and pin gear.***

2.5.2. Mechanism assembly

This sub-assembly consists of the pin sub-assembly, motor, plastic nozzle piece, sector shaped gear, 56 teeth gear, 64 teeth gear, and a pair of M2.5 screws. It's designed to represent the reduction gear mechanism of the gears and the vertical motion of the plastic nozzle piece. The internal frame is set as the default constraint with the 64 teeth gear, 56 teeth gear and sector shaped gear set to pin constraints with axis alignment as the axis for the holes in the gears and protrusions on the internal frame with the back surfaces of the gears coincident to the internal frame top surface. The gear mechanism in the applications portal was used to allow for transfer of rotational motion between gears with the 12 teeth motor gear set to the 56 teeth gear, 56 teeth gear set to the 64 teeth gear and the sector shaped gear set to the 64 teeth gear. The plastic nozzle piece is coincident to the sector shaped gear protrusion with a sketch line used to represent the vertical translation of the plastic nozzle piece. The motor is coincident to the hole on the internal frame with the holes on the motor oriented to the holes on the frame for the M2.5 screws.

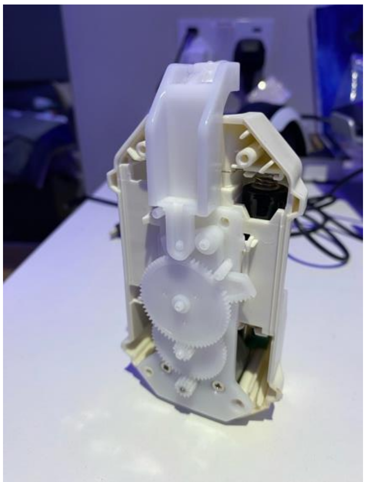

Figure 18: ***image showing photograph of gear mechanism on the top frame***

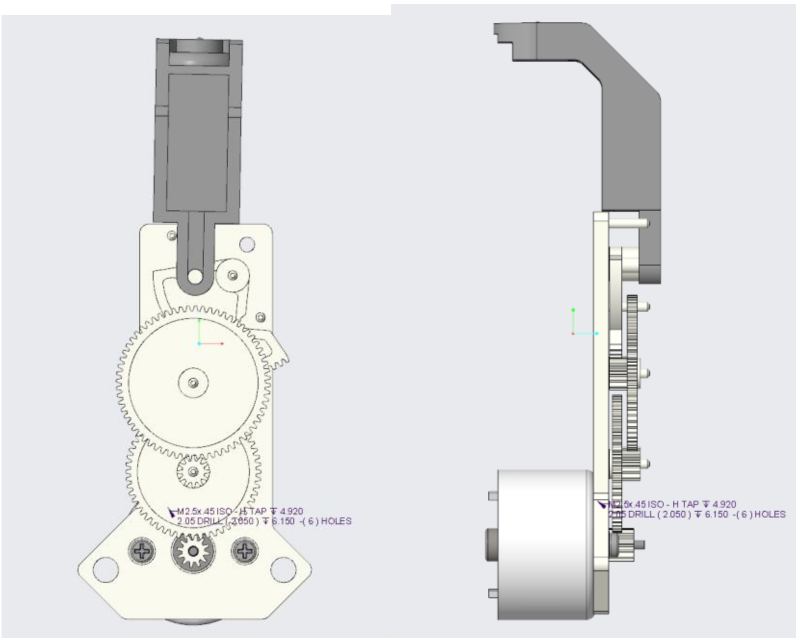

Figure 19: ***image showing model design of gear mechanism from the front and side profile.***

2.5.3. Top assembly

This sub-assembly consists of the fragrance capsule with its top, the top frame, circuit board and a pair of AA batteries shown in figure 20. The top frame is set as the default constraint for the assembly with the two batteries and fragrance capsule coincident to the hollowed sections on the top frame. The circuit board is parallel to the hollowed surfaces of the top frame with a clearance of 1 mm from each surface.

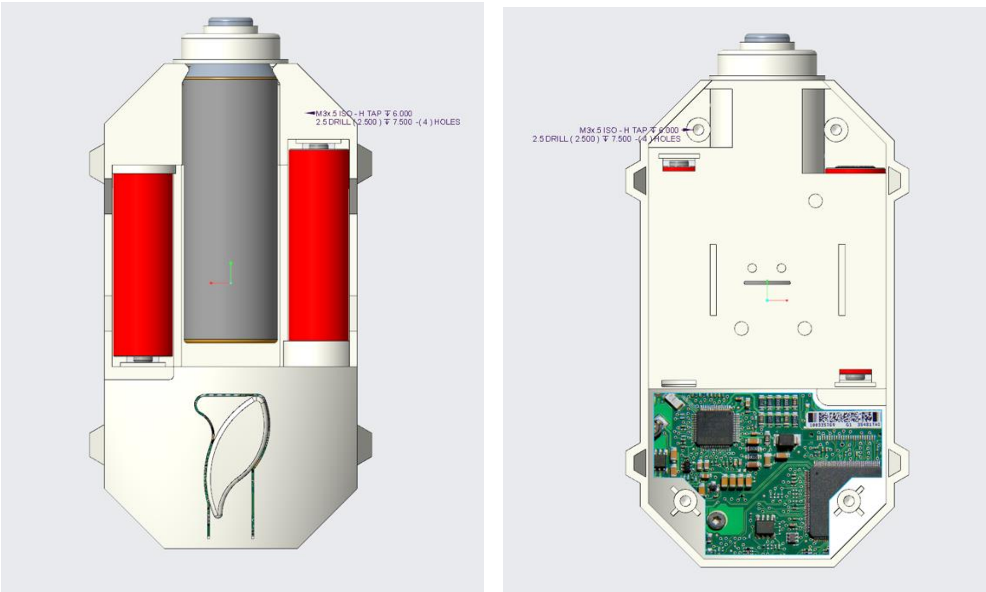

Figure 20: ***model design of top sub-assembly from the front and back face.***

2.5.4. Semi-complete assembly

This sub assembly consists of the top assembly, mechanism assembly, 4 M3 screws, and the bottom frame and illustrates the entire air freshener without the top cover. The internal frame is coincident to the parallel plates on the top frame and oriented using the protrusions on the top frame. The M3 screws are coincident to holes on the bottom frame to complete the back surface shown in figure 21.

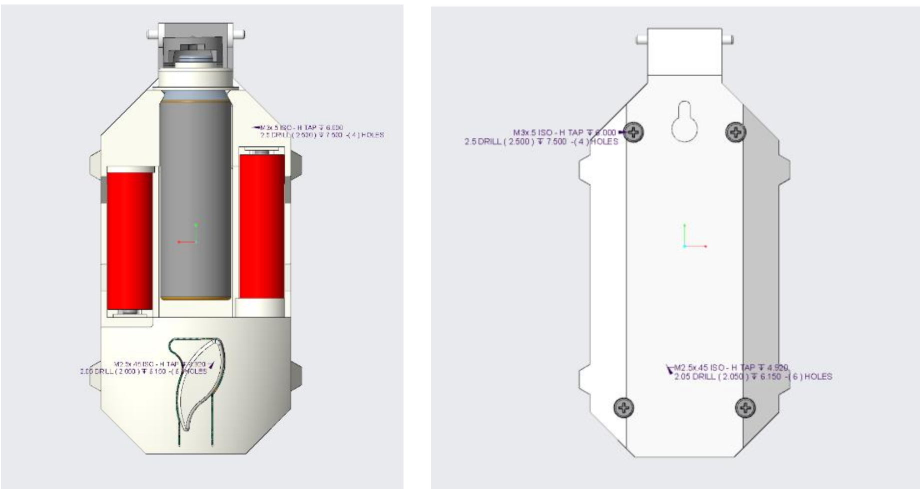

Figure 21: ***images showing model design of the sub assembly from the front and back face.***

2.6. Complete assembly

This section shows the complete assembly with the top cover included represented in figure 22 below with the top cover open and partially closed.

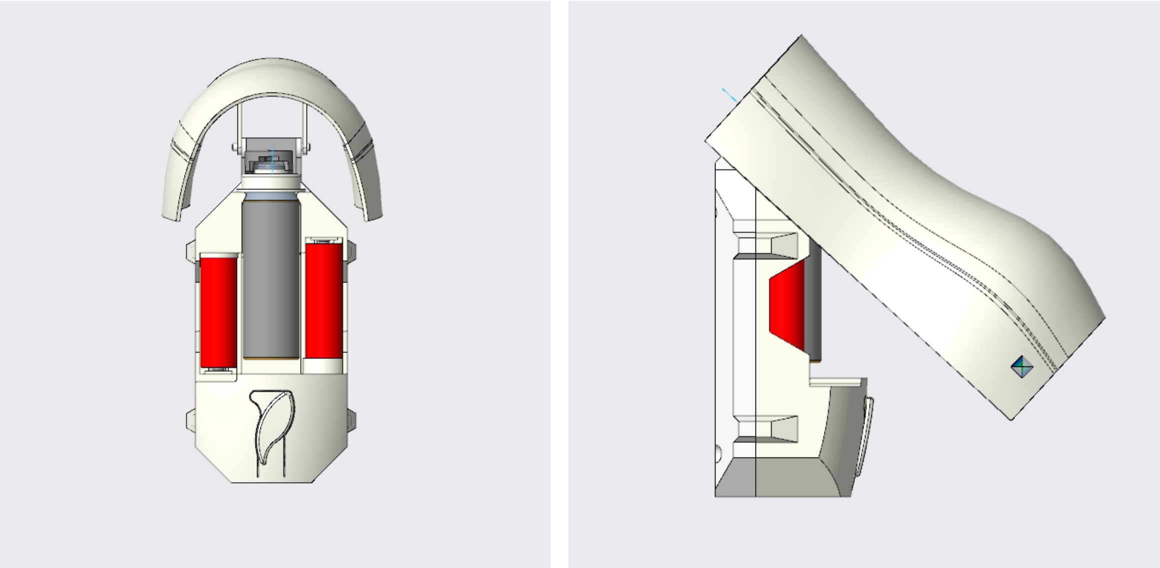

Figure 23: final assembly with top cover open and partially closed.

3. Bill of materials

Item No.	Name	assembly	Material	Quantity
1	M2.5 x 0.45 screw	Mechanism assembly	Stainless steel	2
2	M3 x 5 mm screw	Semi-complete assembly	Stainless steel	4
3	Motor	Mechanism assembly	Aluminium	1
4	Pin	Pin assembly	Stainless steel	1
5	Bottom frame	Semi-complete assembly	ABS plastic	1
6	Top frame	Top assembly	ABS plastic	1
7	Internal frame	Mechanism assembly	ABS plastic	1
8	64 teeth gear	Mechanism assembly	ABS plastic	1
9	56 teeth gear	Mechanism assembly	ABS plastic	1
10	12 teeth motor gear	Pin assembly	ABS plastic	1
11	Sector shaped gear	Mechanism assembly	ABS plastic	1
12	Plastic nozzle piece	Semi-complete assembly	ABS plastic	1
13	capsule	Capsule assembly	Polypropylene	1
14	Capsule top	Capsule assembly	ABS plastic	1
15	Batteries	Top assembly	Polypropylene	2
16	Electric circuit	Top assembly	Pre-peg material	1
17	Top cover	Main assembly	ABS plastic	1

4. References

Reaves, N., 2021. *Creo Parametric Tip: Information About Bill of Materials (BOM) Report Formats*. [online] Rand 3D: Insights from Within. Available at:<https://blogs.rand.com/rand3d/2020/05/creoparametric-tip-information-about-bill-of-materials-bom report-formats.html> [Accessed 11 December 2024].

5. 2D technical drawings.